

Predictive Modeling for Course Demand and Revenue Forecasting on EduPro

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Code: github.com/Nimra-Kulsum/edupro-demand-forecasting | Live dashboard:
<https://edupro-demand-forecasting-szfykjkmf8mtpisvj3cpz.streamlit.app/>

Abstract

Online learning platforms must decide which courses to launch, how to price them and which instructors to onboard, yet EduPro currently relies on historical intuition. This study builds predictive models for course-level enrollments and revenue, and aggregates them to category level, using 60 courses and 10,000 transactions recorded between 2025-01-01 and 2025-12-30. Five regression models (Linear, Ridge, Lasso, Random Forest and Gradient Boosting) were compared on a time-based split: course attributes and past performance up to 2025-09-12 were used to predict demand in the following 109 days. The best enrollment model (Ridge) reached a cross-validated R^2 of -0.45 and the best revenue model (Random Forest) reached 0.97, which we classify as very weak (little better than predicting the average) and strong explanatory power respectively. Key drivers of demand for new courses were: Course Duration, Course Level, Course Price. The models are delivered through an interactive Streamlit dashboard.

1. Introduction

Decisions about launching courses, adjusting prices and onboarding instructors all depend on knowing which courses will attract learners and revenue. Traditional reporting describes what already happened; planning needs forecasts. This project moves EduPro from reactive reporting to proactive planning by answering one question: *which courses will attract enrollments and generate revenue in the future?*

The objectives are to (1) predict enrollment count per course, (2) predict revenue per course and by category, (3) identify the factors that drive demand, and (4) translate the results into business recommendations delivered through a decision-support dashboard.

2. Data

The workbook contains four sheets; this study uses Courses (category, type, level, price, duration, rating), Teachers (expertise, years of experience, rating) and Transactions (course, date, amount). After removing duplicate transactions, records with missing dates or amounts, and transactions for unknown courses, 10,000 of 10,000 transactions and 60 courses remained. Missing values in course and instructor attributes were filled with the median (numeric) or an 'Unknown' label (categorical) inside the modeling pipeline, so no information leaks from the test data.

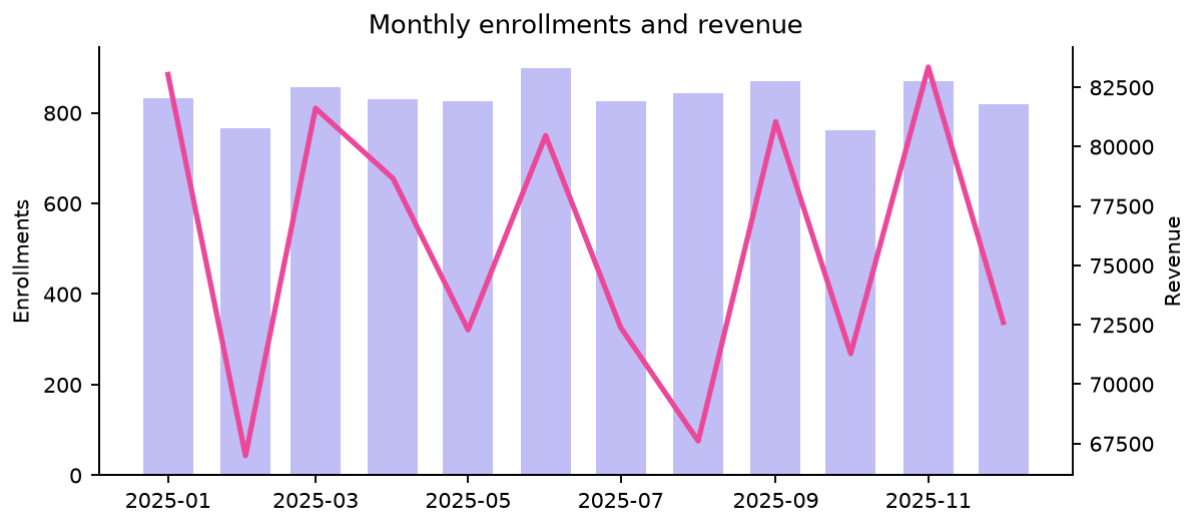


Figure 1. Monthly enrollments (bars) and revenue (line).

3. Methodology

3.1 Target construction (avoiding leakage)

Transactions were split by time. Everything on or before 2025-09-12 is the **history window**; the final 109 days are the **forecast window**. For each course the targets are its enrollments and revenue in the forecast window. Historical features (past enrollments, past average monthly revenue, revenue per enrollment) are computed only from the history window, so the model never sees the outcome it predicts.

3.2 Feature engineering

- **Course features:** price band (low/medium/high), duration bucket, rating tier, level encoding (beginner=0, intermediate=1, advanced=2).
- **Instructor features:** experience bucket, teacher rating, and an expertise-category match score (1.0 exact, 0.7 partial, 0.5 keyword overlap, 0 none).
- **Historical performance:** past enrollment count, past average monthly revenue and revenue per enrollment.
- **Redundancy removal:** numeric features with absolute correlation above 0.9 were dropped automatically (removed from the full model: YearsOfExperience, TeacherRating, ExpertiseMatch, PastAvgMonthlyRevenue, PastRevenue, RevenuePerEnrollment).

3.3 Two model families

A **launch model** uses only course and instructor attributes, so it can score a brand-new course that has no history. A **full model** adds historical performance and is used to forecast existing courses and categories.

3.4 Models and evaluation

Baselines were Linear, Ridge and Lasso regression; advanced models were Random Forest and Gradient Boosting. Categorical variables were one-hot encoded and numeric variables standardized. Each model was evaluated on a 25% hold-out set using MAE (average error), RMSE (penalizes large errors) and R2 (variance explained), and with 5-fold cross-validation (CV R2), which is more reliable with few courses. The model with the highest CV R2 was selected for each target.

4. Exploratory Data Analysis

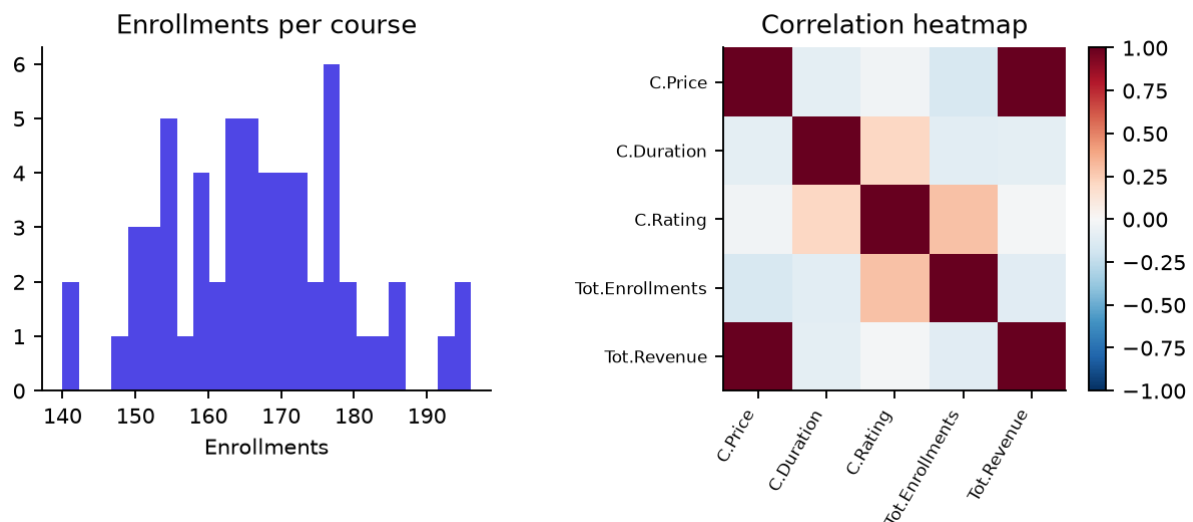


Figure 2. Distribution of enrollments per course and correlation heatmap of numeric variables.

Relationship of course attributes with total enrollments: Course Price has a slight negative relationship with enrollments (Spearman rho = -0.16); Course Rating has a slight positive relationship with enrollments (Spearman rho = 0.28); Course Duration shows no clear monotonic relationship with enrollments (Spearman rho = -0.07).

Price band	Courses	Avg enrollments	Avg revenue
Low	38	168.5	0
Medium	2	171.0	3,306
High	20	162.8	45,236

Course level	Courses	Avg enrollments	Avg revenue
Beginner	21	170.1	17,572
Intermediate	18	164.0	14,681
Advanced	21	165.5	13,240

Courses in the **Medium** price band attract the highest average enrollments, and **Beginner**-level courses lead among levels. By total historical revenue, the leading category is **Artificial Intelligence** (202,751) and the smallest is **Marketing** (0).

5. Results

5.1 Model comparison

Tables 1 and 2 report all models. For enrollments, the best full model is **Ridge** (CV R2 = -0.451, MAE = 6.3, RMSE = 8.5); the best launch model is **Random Forest** (CV R2 = -0.404). For revenue, the best full model is **Random Forest** (CV R2 = 0.971, MAE = 973); the best launch model is **Random Forest** (CV R2 = 0.968). The simple Linear Regression baseline scored a CV R2 of -1.326 for enrollments, so the best model improves on the baseline (+0.875).

Table 1. Enrollment models

Feature set	Model	MAE	RMSE	R2 (test)	CV R2
Full (with history)	Ridge	6.3	8.5	-0.599	-0.451 +/- 0.11
Full (with history)	Random Forest	6.3	8.1	-0.457	-0.451 +/- 0.03
Full (with history)	Lasso	6.9	9.6	-1.012	-0.642 +/- 0.16
Full (with history)	Gradient Boosting	9.3	10.8	-1.559	-0.966 +/- 0.40
Full (with history)	Linear Regression	8.8	12.2	-2.278	-1.326 +/- 0.40
Launch (new course)	Random Forest	5.5	7.6	-0.278	-0.404 +/- 0.10
Launch (new course)	Ridge	6.0	8.1	-0.450	-0.417 +/- 0.10
Launch (new course)	Lasso	6.3	8.8	-0.687	-0.594 +/- 0.10
Launch (new course)	Gradient Boosting	7.5	9.0	-0.766	-0.962 +/- 0.57
Launch (new course)	Linear Regression	8.1	10.7	-1.495	-1.200 +/- 0.27

Table 2. Revenue models

Feature set	Model	MAE	RMSE	R2 (test)	CV R2
Full (with history)	Random Forest	972.5	1,791.7	0.956	0.971 +/- 0.01
Full (with history)	Gradient Boosting	771.2	1,292.4	0.977	0.968 +/- 0.02
Full (with history)	Lasso	811.8	1,220.5	0.980	0.939 +/- 0.06
Full (with history)	Linear Regression	814.1	1,222.6	0.980	0.938 +/- 0.06
Full (with history)	Ridge	1,474.9	1,987.8	0.946	0.849 +/- 0.22
Launch (new course)	Random Forest	908.5	1,752.5	0.958	0.968 +/- 0.01
Launch (new course)	Gradient Boosting	929.8	1,617.1	0.964	0.961 +/- 0.02
Launch (new course)	Lasso	824.3	1,255.2	0.978	0.946 +/- 0.04
Launch (new course)	Linear Regression	826.2	1,256.4	0.978	0.946 +/- 0.04
Launch (new course)	Ridge	1,426.5	1,960.2	0.947	0.851 +/- 0.22

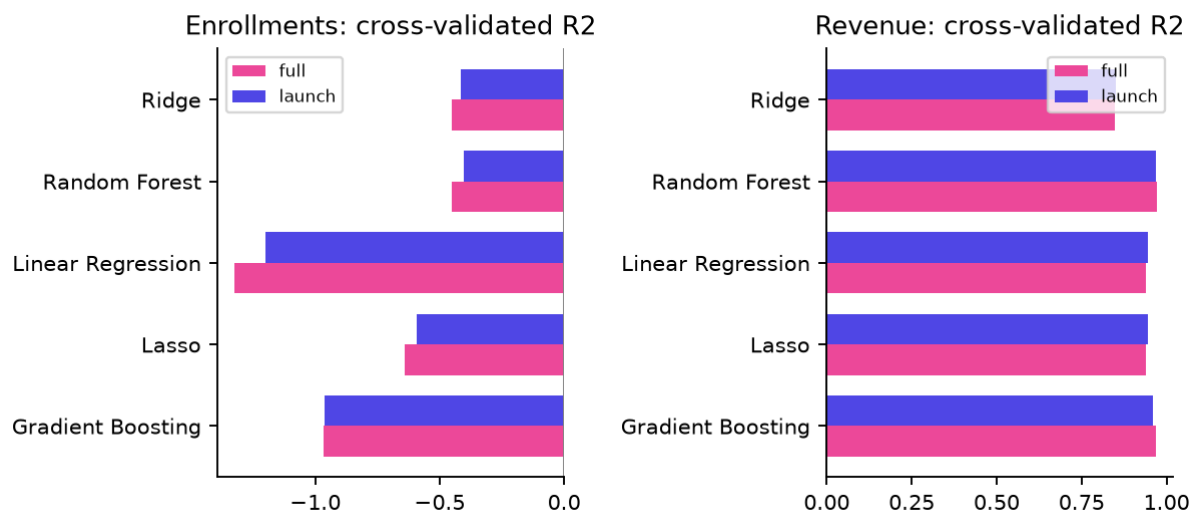


Figure 3. Cross-validated R2 by model and feature set.

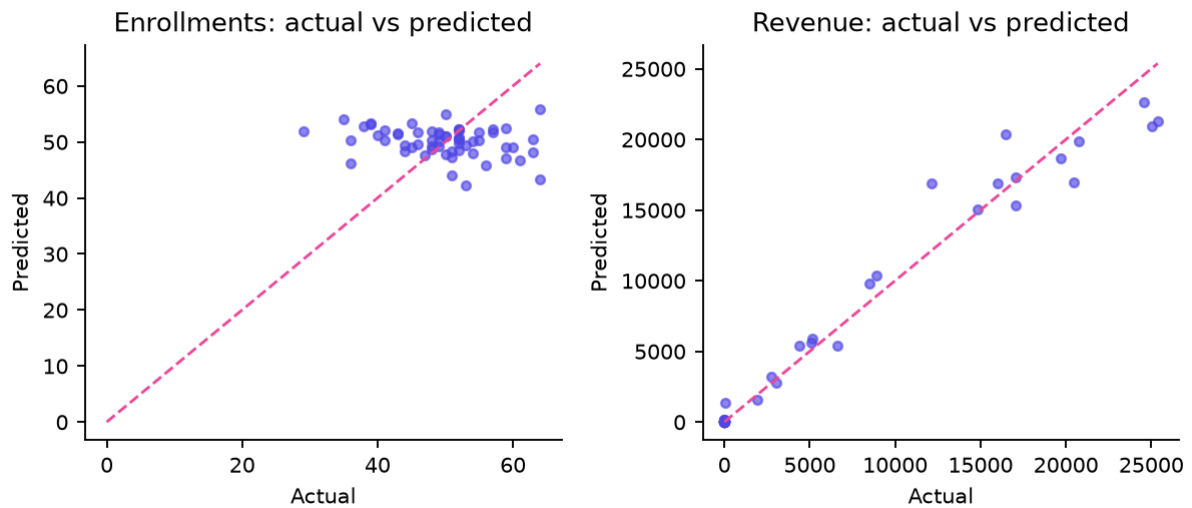


Figure 4. Out-of-fold predictions versus actual values for the full models.

5.2 Category-level forecast

Summing course-level predictions gives a category forecast for the next 109 days. The category with the highest predicted revenue is **Artificial Intelligence** (58,210) and the lowest is **Marketing** (45).

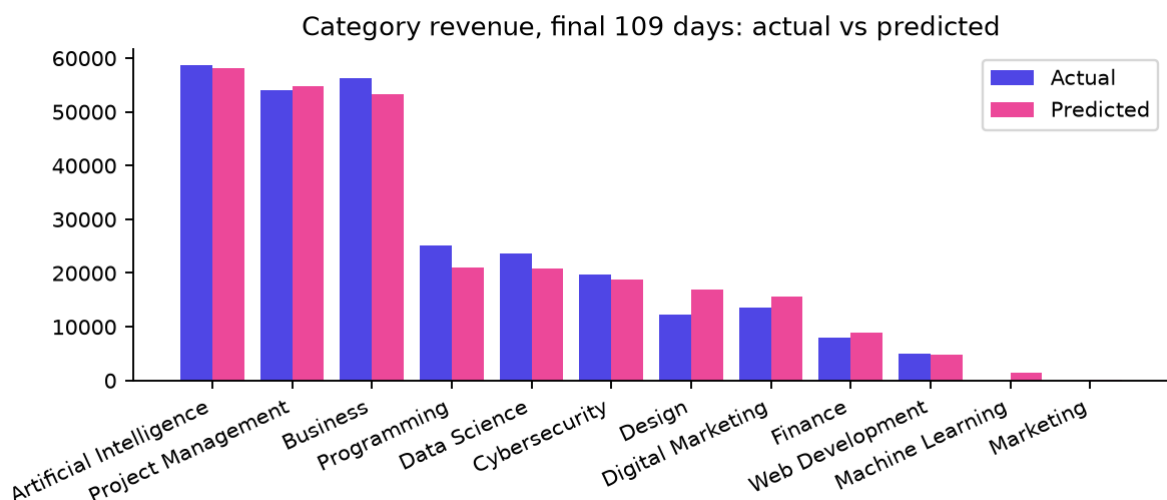


Figure 5. Actual versus predicted revenue by category (out-of-fold predictions).

6. Demand Drivers

Importance was measured by shuffling each group of related features on held-out courses and recording the fall in R^2 ; a larger fall means the model relies on that factor more. Values near zero mean the factor adds nothing beyond noise.

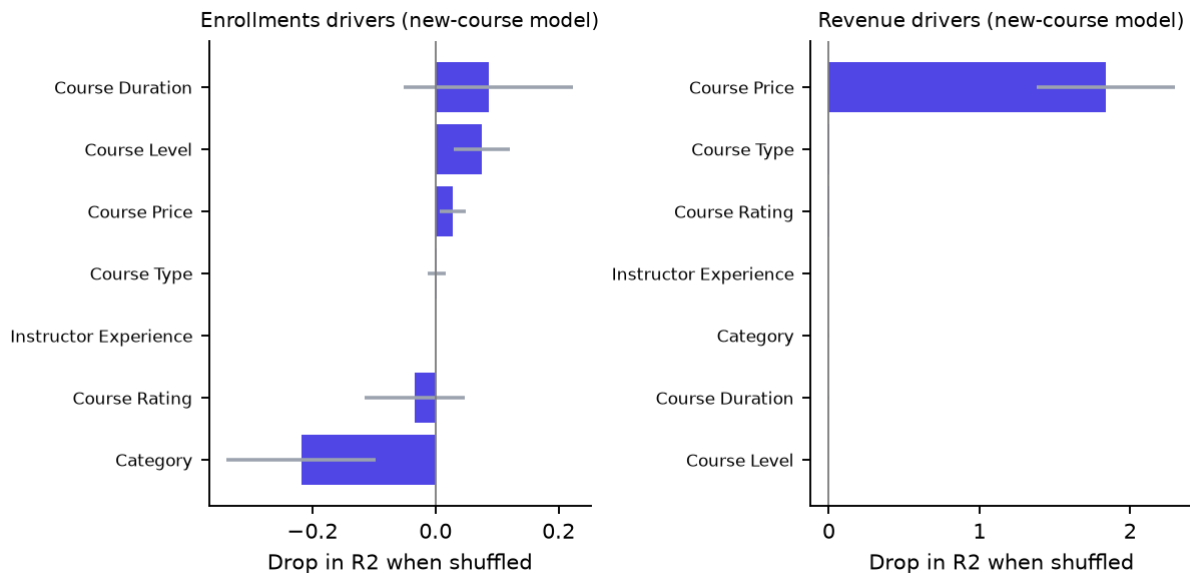


Figure 6. Grouped permutation importance for the new-course (launch) models.

For enrollments the factors that matter most are: **Course Duration, Course Level, Course Price**. For revenue they are: **Course Price**.

7. Recommendations

- **Screen launches with the Demand Predictor.** Test price, level and instructor profile for a proposed course before committing budget; treat results as a ranking aid.
- **Prioritise categories by forecast revenue.** Invest first in **Artificial Intelligence**; review positioning for **Marketing**.
- **Use price bands deliberately.** The **Medium** band currently attracts the most enrollments per course; test price changes gradually and measure the effect.
- **Focus on the drivers.** Work on: Course Duration, Course Level, Course Price.
- **Re-train quarterly** as new transactions arrive and track forecast versus actual to detect drift.

8. Limitations and Future Work

- Model accuracy is bounded by the signal in the available attributes; the best CV R2 values (-0.45 for enrollments, 0.97 for revenue) are strong.
- The dataset has 60 courses, so test-set metrics are noisy; cross-validation was used to compensate.
- Marketing activity, promotions, seasonality and competitor pricing are not in the data and may strongly influence demand.
- Predictions are statistical estimates, not guarantees; the price-sensitivity curve in the dashboard is directional, not causal.
- Future work: add time-series models (e.g. Prophet), learner-level features, and A/B tests of price changes.

9. Conclusion

This project replaces intuition-based planning with quantified, reproducible forecasts. Using a leakage-free time-based design, five models were compared for enrollments and revenue; the selected models offer very weak (little better than predicting the average) (enrollments) and strong (revenue) explanatory power, and the dashboard turns them into a practical tool for launch, pricing and resource decisions. The honest reporting of accuracy and limitations is intended to support careful, evidence-based use.